```
import express from "express"; // import Express framework
import bodyParser from "body-parser"; // to read form data
import pg from "pg"; // PostgreSQL library

const app = express(); // create app
const port = 3000; // server port

// Database connection
const db = new pg.Client({
  user: "postgres",        // DB username
  host: "localhost",       // DB host
  database: "permalist",   // DB name
  password: "12345",       // DB password
  port: 5432,              // DB port
});
db.connect(); // connect to database

app.use(bodyParser.urlencoded({ extended: true })); // read form data
app.use(express.static("public")); // serve CSS/images
app.get("/", async (req, res) => {
  try {
    const result = await db.query("SELECT * FROM items ORDER BY id ASC"); // get
all items
    const items = result.rows; // store rows

    res.render("index.ejs", {   // send data to EJS
      listTitle: "Today",       // page title
      listItems: items,         // list of items
    });
  } catch (err) {
    console.log(err); // show error
  }
});
app.post("/add", async (req, res) => {
  const item = req.body.newItem; // get input value

  try {
    await db.query("INSERT INTO items (title) VALUES ($1)", [item]); // insert
into DB
    res.redirect("/"); // refresh page
  } catch (err) {
    console.log(err);
  }
});
```

```
app.post("/edit", async (req, res) => {
  const item = req.body.updatedItemTitle; // new text
  const id = req.body.updatedItemId; // item id

  try {
    await db.query(
      "UPDATE items SET title = ($1) WHERE id = $2", // update query
      [item, id]
    );
    res.redirect("/");
  } catch (err) {
    console.log(err);
  }
});
app.post("/delete", async (req, res) => {
  const id = req.body.deleteItemId; // get id

  try {
    await db.query("DELETE FROM items WHERE id = $1", [id]); // delete
    res.redirect("/");
  } catch (err) {
    console.log(err);
  }
});
```

EJS File

```
<%- include('partials/header.ejs'); -%> <!-- include header -->

<div class="box" id="heading">
  <h1><%= listTitle %></h1> <!-- show title -->
</div>

<div class="box">

  <% for(let item of listItems){ %> <!-- loop through items -->

    <div class="item">

      <!-- DELETE FORM -->
      <form action="/delete" method="post">
        <input type="checkbox"
               onchange="this.form.submit()"  <!-- auto submit -->
```

```
            name="deleteItemId"
            value="<%= item.id %>"> <!-- send id -->
    </form>

    <!-- SHOW ITEM TEXT -->
    <p id="title<%=item.id%>">
      <%= item.title %>
    </p>

    <!-- EDIT FORM -->
    <form class="edit" action="/edit" method="post">
      <input type="hidden"
            name="updatedItemId"
            value="<%= item.id %>"> <!-- hidden id -->

      <input id="input<%=item.id%>"
            type="text"
            name="updatedItemTitle"
            value="<%= item.title %>"
            hidden="true" /> <!-- hidden input -->

      <button id="done<%=item.id%>" hidden>
        ✓ <!-- save button -->
      </button>
    </form>

    <!-- EDIT BUTTON -->
    <button id="edit<%=item.id%>"
           onclick="handler('<%=item.id%>')">
      ✎ <!-- click to edit -->
    </button>

  </div>

<% } %>

<!-- ADD NEW ITEM -->
<form class="item" action="/add" method="post">
  <input type="text"
         name="newItem"
         placeholder="New Item" /> <!-- input -->

  <button class="add" type="submit">+</button> <!-- add -->
</form>
```

```
</div>

<script>
function handler(id) {
  // hide text
  document.getElementById("title" + id).setAttribute("hidden", true);

  // hide edit button
  document.getElementById("edit" + id).setAttribute("hidden", true);

  // show input box
  document.getElementById("input" + id).removeAttribute("hidden");

  // show done button
  document.getElementById("done" + id).removeAttribute("hidden");
}
</script>

<%- include('partials/footer.ejs'); -%> <!-- include footer -->
```

Header

```
<!DOCTYPE html>

<html>

<head>

  <title>Permalist</title> <!-- page title -->

  <link rel="stylesheet" href="styles/main.css"> <!-- CSS -->

</head>

<body>

<main>
```

Footer

```
</main>

<footer>
  Copyright © <%= new Date().getFullYear() %> <!-- dynamic year -->
</footer>

</body>
</html>
```

CSS FILE

```
/* Page background and font */
html {
  background-color: #e4e9fd; /* light background color */
  background-image: -webkit-linear-gradient(65deg, #a683e3 50%, #e4e9fd 50%); /* gradient design */
  min-height: 1000px; /* minimum page height */
  font-family: "helvetica neue"; /* font style */
}

/* Body layout */
body {
  display: flex; /* use flexbox */
  flex-direction: column; /* vertical layout */
  justify-content: stretch; /* stretch content */
  min-height: 95vh; /* full screen height */
}

/* Main content area */
```

```
main {
  text-align: center; /* center text */
  flex: 1 0 auto; /* flexible size */
}

/* Heading style */
h1 {
  color: #fff; /* white text */
  padding: 10px; /* spacing */
}

/* Main container box */
.box {
  max-width: 400px; /* max width */
  margin: 50px auto; /* center horizontally */
  background: white; /* white background */
  border-radius: 5px; /* rounded corners */
  box-shadow: 5px 5px 15px -5px rgba(0, 0, 0, 0.3); /* shadow */
}

/* Header box */
#heading {
  background-color: #a683e3; /* purple background */
  text-align: center; /* center text */
}

/* Edit form layout */
form.edit {
  display: flex; /* align items in row */
```

```
}

/* Each item row */
.item {
  min-height: 70px; /* height */
  display: flex; /* flex layout */
  align-items: center; /* center vertically */
  border-bottom: 1px solid #f1f1f1; /* bottom border */
}

/* Remove border from last item */
.item:last-child {
  border-bottom: 0;
}

/* Checkbox spacing */
input[type="checkbox"] {
  margin: 20px; /* space around */
  margin-left: 0; /* no left space */
}

/* Text style */
p {
  margin: 0; /* remove default margin */
  padding: 20px 0; /* vertical spacing */
  font-size: 20px; /* text size */
  color: #00204a; /* dark blue color */
}
```

```
/* Form spacing */
form {
  text-align: center; /* center content */
  margin-left: 20px; /* left space */
}

/* Add button (+) */
button.add {
  min-height: 50px; /* height */
  width: 50px; /* width */
  border-radius: 50%; /* circle shape */
  border-color: transparent; /* no border */
  background-color: #a683e3; /* purple */
  color: #fff; /* white text */
  font-size: 30px; /* large + */
  border-width: 0; /* remove border */
}

/* Edit button (pencil/check) */
button.edit {
  margin-left: 0px; /* no left margin */
  margin-right: 20px; /* right space */
  border: none; /* no border */
  background: none; /* transparent */
}

/* Icon size */
.icon {
  height: 20px; /* small icons */
```

```
}

/* Text input field */
input[type="text"] {
  text-align: left; /* align text */
  height: 60px; /* height */
  top: 10px; /* position */
  border: none; /* remove border */
  background: transparent; /* no background */
  font-size: 20px; /* text size */
  font-weight: 200; /* light text */
  width: 80%; /* width */
}

/* Input focus effect */
input[type="text"]:focus {
  outline: none; /* remove outline */
  box-shadow: inset 0 -3px 0 0 #a683e3; /* bottom border effect */
}

/* Placeholder text */
::placeholder {
  color: lightgray; /* light color */
  opacity: 1; /* visible */
}

/* Footer */
footer {
  flex: 0 0 auto; /* fixed size */
```

```
  color: white; /* white text */
  text-align: center; /* center text */
}
```